

16. Regard $\mathbb{Q}$, the set of all rational numbers, as a metric space, with $d(p, q) = |p - q|$. Let E be the set of all $p \in \mathbb{Q}$ such that $2 < p^2 < 3$. Show that E is closed and bounded in $\mathbb{Q}$ but that E is not compact. Is E open in $\mathbb{Q}$?

First we will show that E is bounded.

Consider the point $q = 1.5$ and the neighbourhood around it of radius 0.5 . We can show that E is contained in that neighbourhood, and is thus bounded. We prove this by proving that any point not contained in the neighbourhood is not contained in the set as below.

If $d(p, q) > 0.5$ then we have

$$|p - 1.5| > 0.5$$

which gives us $p - 1.5 > 0.5$ or $p - 1.5 < -0.5$.

Then we must have $p > 2$ or $p < 1$ and thus $p^2 < 1$ or $p^2 > 4$. This gives us $p \notin E$. Thus, we've shown that E is bounded. Now we will show that E is closed by proving that any point that is not in E cannot be a limit point of E .

If $p \notin E$ then we must have $p^2 < 2$ or $p^2 > 3$. In each of these cases we can construct a neighbourhood around p that does not include any points of E .

By the construction in chapter 1 to prove that the rationals do not have the least upper bound property, we know that there exists a positive rational r such that for any p with $p^2 < 2$ we have $(p + r)^2 < 2$. Specifically,

$$r = \min\left(\frac{2 - p^2}{2p + 1}, 1\right)$$

gives us this property. Moreover, since r is positive and the square is an increasing function for positive reals, this gives us $q^2 < 2$ for any $q \in (p - r, p + r)$. Thus $N_r(p)$ contains no points of E making p not a limit point of E for any p with $p^2 < 2$.

We have a similar proof to show that p with $p^2 > 3$ also cannot be a limit point of E . Particularly, we use a similar construction for r such that the $N_r(p)$ contains no points of E whenever $p^2 > 3$.

Consider

$$r = \min\left(\frac{p^2 - 3}{2p + 1}, 1\right)$$

for some p with $p^2 > 3$. Then

$$(p - r)^2 = p^2 - r(2p - r) > p^2 - r(2p + 1)$$

since $-r < 1$. Substituting r gives us

$$(p - r)^2 > p^2 - \left(\frac{p^2 - 3}{2p + 1}\right)(2p + 1) = p^2 - (p^2 - 3) = 3.$$

That is

$$(p - r)^2 > 3.$$

Again, since r is positive, and the square function is increasing, this gives us this gives us $q^2 > 3$ for any $q \in (p - r, p + r)$. Thus $N_r(p)$ contains no points of E making p not a limit point of E for any p with $p^2 > 3$.

Since we've exhausted all the cases for rationals not in E (there is no rational p with $p^2 = 2$ or $p^2 = 3$) and shown all of them not to be limit points, we've shown that all the limit points of E must be in E , and thus, E must be closed.

Finally, now we will show that E is compact. Consider the cover of E given by $\{G_p\}$ where $G_p = (p - r_p, p + r_p)$ such that $[p - r_p, p + r_p] \subseteq E$ for all $p \in E$. Note that such an r_p exists for each $p \in E$ by a similar construction to that which we used earlier.

For any finite subset of $\{G_p\}$, say S , there must be some maximum $p + r_p$ chosen among all the G_p in S . Then that maximum $p + r \notin \bigcup_{G_p \in S} G_p$ but $p + r \in E$ making S not a finite subcover, and making E not compact. ■

17. Let E be the set of all $x \in [0, 1]$ whose decimal expansion contains only the digits 4 and 7. Is E countable? Is E dense in $[0, 1]$? Is E compact? Is E perfect?